

# One clue, two cities?

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Use heights 1, 2, 3 once in each row and once in each column. A number outside counts towers seen from that end; taller towers hide shorter ones.

## 1. Build a city.

First make any legal city with your towers. Then add the clue shown here: three towers must be visible down the first column.

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## 2. Find every city with that clue.

Record two different cities above. Are there more? Keep the board in the same orientation.

**Explain why your list is complete. What can go in the top row?**

**Hint after trying:** The first column is forced. Try each possible order of the two remaining heights in the top row.

Build first. Explain by pointing, talking, or drawing.

# How little information is enough?

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## 3. Make just one city possible.

Keep the top clue 3. Add a right-side clue of 2 to the **middle row**. Which of your two cities survives? Show why.

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## 4. Is either clue unnecessary?

Cover the right clue: your first page gives two answers. Now cover only the top clue. Find two cities that keep the right clue.

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2

**Would any single outside clue ever force an entire 3-by-3 city?**

**Try a reason covering all clues:** Keep the row or column that a clue sees fixed. Can you swap two other rows or columns?

Build first. Explain by pointing, talking, or drawing.